POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the Java architecture and significance of class path. Differentiate between error and exception. 8
   b) Define Package. Explain the types of access modifiers in Java. 7
2. a) Define interface. Write Java program that override methods (getUserValue() and displayUserDetail()), where getUserValue() is defined to get user detail(name, address, age) and displayUserDetail() to display user detail. 8
   b) Define subclass and super class. How can we achieve multiple inheritance in Java? Explain. 7

OR

Differentiate between Java abstract class and interface with a sample program

3. a) Write a program to create a Frame that has two TextField components, one Label, and a Button. When the user clicks on the button, calculate the sum of the values entered in the first and second TextField and display the result on the third Label. 8
   b) What are the pros and cons of JavaFX. Explain any two JavaFX layout managers. 7
4. a) Create a TCP client server application where the client sends a string and the server responds by echoing the same string in upper case. 8
   b) What is JavaMail API? Explain type of protocol used in JavaMail API. 7

OR

Define RMI stubs and skeleton. Create an RMI application where a client can remotely invoke a method that sends the sum of any two given integers.

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5. a) A table tbl_student consists of id, name and program. Write a program that asks users to enter a program name and displays all the students records enrolled in that program. 8

   b) Differentiate between GET and POST methods. Write java program that depict Sessions and Cookies. 7

6. a) Explain life cycle methods of servlet. Write a servlet program that accept input string from user through form and then check if it is palindrome or not. 8

   b) Why do we need design patterns? Explain Singleton Pattern and Factory Method Pattern in detail. 7

7. Write short notes on: **(Any two)** 2x

   a) SQL Injection

   b) ORM

   c) CORBA

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advance Programming with Java

Semester: Spring

Year : 2024
Full Marks:100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define ClassPath? Write a program with custom exception handler that handle arithmetic exception. 7

   b) Explain the architecture of Java, including the role of the JVM, JRE, and JDK. 8

OR

Define collection framework in java. Explain the different components of java collection framework.

2. a) What are the implications in Java when defining a variable, method, and class as final? Explain the various access modifiers in Java and their significance. 8

   b) Define abstraction. Write Java program that override methods (getUserValue() and displayUserDetail()). Where getUserValue() is defined to get user detail(name, address, age) and displayUserDetail() to display user detail. 7

3. a) Pros and Cons of JavaFx. Create GUI application with two buttons "RED" and "Green". When clicking "RED" button change background of "GREEN" button to red and clicking "GREEN" button change background of "RED" button to green. Also, console button clicked message. 8

   b) Define stage and scene in JavaFX. Explain any two JavaFX layout managers. 7

4. a) Explain RMI architecture. Develop RMI application where client request to identify given number is Odd or Even. 8

   b) Write a TCP client server program, where a client sends an integer value and server responds by sending the squared value of that integer. 7

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5. a) Differentiate between Statement and PreparedStatement. Consider db_college and tbl_student as database name and table respectively where a tbl_student has columns name, faculty and batch. Write Console application with CREATE, READ, UPDATE and DELETE queries only using Prepared Statement. 8

b) What is a JDBC driver? Explain the different JDBC drive types and can figuration 7

OR

Write all steps to connect Java Application with any Database. Insert student detail (roll,name,address,age) into table "student_tbl"

6. a) Describe the life cycle of a servlet. Explain the use of doGet() and doPost() methods of the Servlet with an example. 7

b) Create a Servlet application to showcase the concept of session and cookies 8

7. Write short notes on: **(Any two)** 2x5

a) Design patterns

b) Over view of ORM

c) RMI vs. CORBA

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the role of the JVM in Java Architecture? Describe the steps to compile and run a Java program. 7

   b) Write a multithreaded program in Java to print even and odd numbers alternatively. 8

OR

What are design patterns? Compare the Singleton pattern with the Factory pattern.

2. a) Define Interface. As we know, there is no room for multiple inheritance using classes. How do you achieve multiple inheritance in java. Explain with Example. 8

   b) What is a constructor in Java? Explain the different types of constructors with examples. 7

3. a) Explain the event handling mechanism in Java. Write a program that uses ActionListener to handle button clicks. 7

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OR

Define stage and scene in JavaFX. Differentiate between Swing and JavaFx.

   b) Compare the different layout managers in Java Swing with examples for each. 8

4. a) Develop RMI application where client input two numbers if both numbers are even then result $(a+b)^2$ otherwise $(a+b)^3$. 8

   b) Write a Java program to implement a client-server application using TCP sockets. Explain the flow of communication. 7

5. a) What are the main components of JDBC architecture? Explain the role of the DriverManager and Connection interface. 7

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b) Write a Java program to establish a connection to a database using JDBC. Demonstrate the use of *Statement* and *PreparedStatement* objects with examples. 8

6. a) What are the differences between static and dynamic web applications? How does a servlet fit into a dynamic web application? 7

b) Discuss the life cycle of a servlet in detail, including the methods init(), service(), and destroy(). 8

7. Write short notes on: **(Any two)** 2×5

a) Handling GET/POST Requests

b) Hibernate

c) Advantages of MVC

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POKHARA UNIVERSITY

Semester: Spring

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define the term Architectural Neutral. Write down the code to handle user defined exception for LowBalance when balance is less then 10 rupees on your mobile. 7

   b) Define multithreading. Explain the techniques to create thread using anonymous class. 8

OR

Write down the advantage of ORM over JDBC. Explain Singleton Pattern and Factory method pattern in detail.

2. a) Explain the concept of inheritance in Java. What are the benefits of using inheritance, and how does it promote code reusability and polymorphism? Provide an example to support your explanation. 8

   b) Differentiate between abstract class and interface? Explain final classes and methods. 7

3. a) Write down the JavaFx code to design a login page as shown in the figure below when user click on login button test the username is "PU" and password is "PU@2025Spring" or not. 8



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b) Why do we need layout management? Explain GridBag layout with suitable example. 7

a) What is the significance of stub and skeleton in RMI? Create a RMI application such that a client sends an integer number to the server and the server return the factorial value of that integer. 8

b) Create a TCP client server application where the client sends a string and the server responds by echoing the same string in upper case. 7

a) A table student consists of id, name and program. Write a program that asks users to enter a program name and displays the total number of students enrolled in that program. 7

OR

What is SQL Injection Attack? Write sample program using prepared statement to prevent SQL Injection Attack.

b) What is package in java? How can you create user–defined package in java? Explain with example. 8

6. a) Differentiate between servlet and JSP. Create a HTML file with two text fields to First name and Last name. Then create a JSP file that reads values from the HTML form and display full name. 8

b) What is cookie? How can you store and read cookies by using servlet. Explain with proper example. 7

7. Write short notes on: (Any two) 2×5

a) Hibernate ORM

b) CORBA

c) SQL Escapes

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a Just-In-Time (JIT) compiler? Explain the architecture of Java and describe the roles of JDK, JRE, and JVM in the execution of a Java program. 7

   b) Define error and exception. Write a program to ask the user to enter two integers and perform division. Catch all relevant exceptions. 8

   OR

   Define collection framework. How do you create dynamic and unique list using collection framework with code example?

2. a) What is polymorphism? Write down the code to achieve compile time and runtime polymorphism in java using method overloading and overriding. 7

   b) Define an abstract class in Java. Explain the use of extends and super keyword with necessary example code. 8

3. a) Define Layout Management. Compare the different types of Swing and JavaFX layout in brief. 7

   b) Differentiate between AWT and Swing. Create suitable GUI using java swing to input first name and last name separately and finally display the full name when user clicks the submit button. 8

4. a) Create a TCP client server application where the client sends a string and the server responds by replying the length of the string. 7

   b) Define RMI stubs and skeleton. Create an RMI application where a client can remotely invoke a method that checks if the number is even or odd. 8

5. a) Explain the different types of JDBC drivers with diagrams. Discuss about SQL Injection and how PreparedStatement in Java overcomes this problem. 7

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b) Define JDBC. Write down the code to authenticate a user from database. 8

6. a) Define JSP and Servlet. Write down the necessary HTML and servlet code to display a username field and a button Greet. When user clicks on Greet button display "Welcome <<username>>" message in the web page. 7

b) What is ORM and how does it differ from JDBC? Also, define a web framework with examples and discuss the Spring framework in detail. 8

OR

What is private constructor? Explain the significance of private constructor in singleton design pattern with necessary example code.

7. Write short notes on: **(Any two)** 2×5

a) Multithreading in Java

b) Lifecycle of servlet

c) Transaction

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